Credit Card Fraud Detection

Imbalanced Classification with Precision-Recall Optimization

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Agenda

1. Business Context
2. EDA
3. Model Results
4. Conclusions

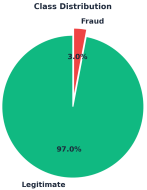
The Fraud Challenge

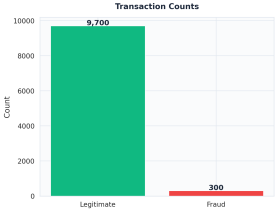
Key Challenges

- **Extreme imbalance**: only 3% of transactions are fraud
- Standard accuracy is misleading (97% by predicting all legitimate)
- Must optimize for **recall** (catch fraud) and **precision** (avoid false alerts)

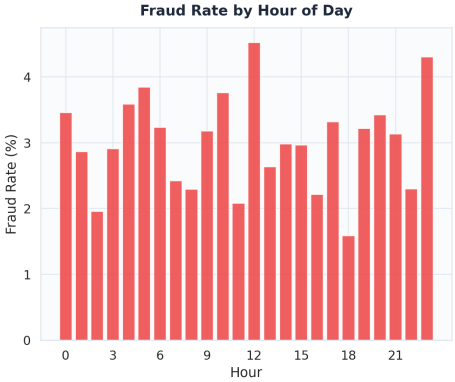
Approach

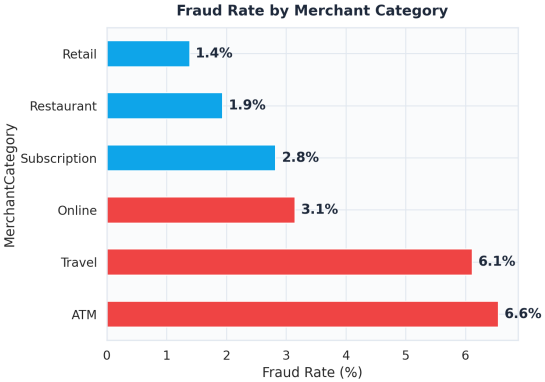
Class-weighted models + threshold optimization + AUC-PR as primary metric.





Fraud Patterns

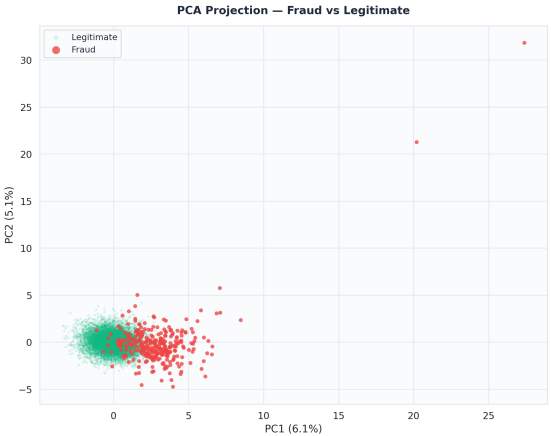


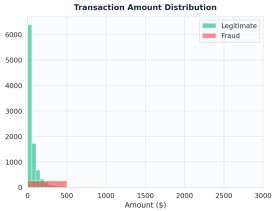


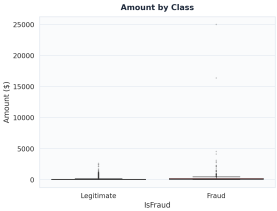
Insights

Fraud rates vary by time of day and merchant category. Online and ATM transactions show elevated fraud risk. International transactions are 2x more likely to be fraudulent.

PCA Projection & Amount Analysis



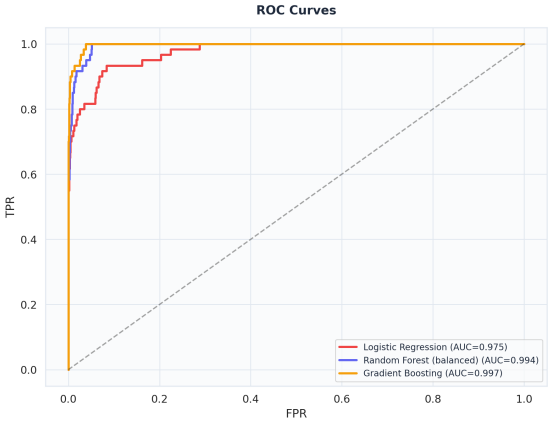


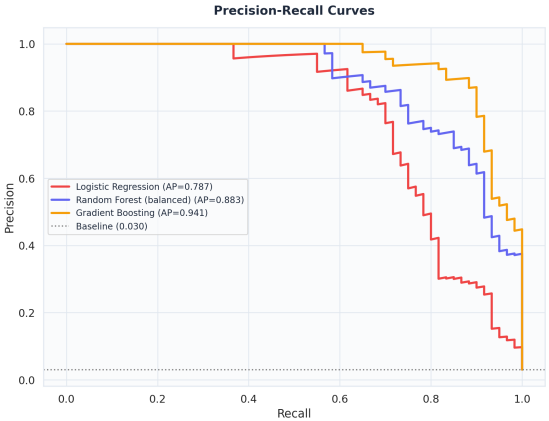


Key Finding

Fraudulent transactions occupy distinct regions in PCA space and tend toward **higher amounts** than legitimate transactions.

ROC & Precision-Recall Curves

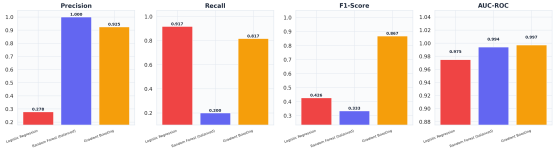


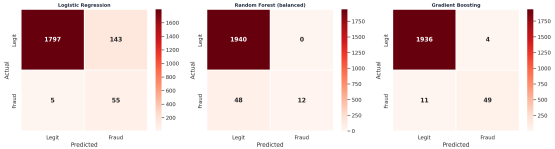


Best Model: Gradient Boosting

AUC-ROC = 0.997 AUC-PR = 0.941 F1 = 0.867 Recall = 82%

Model Comparison & Confusion Matrices

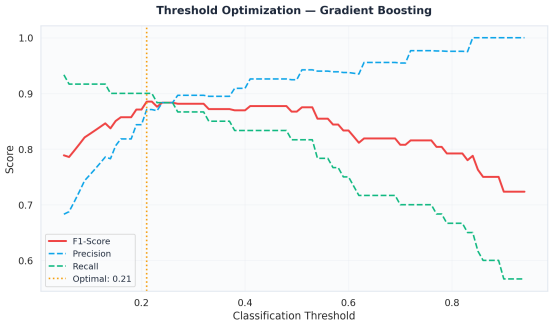


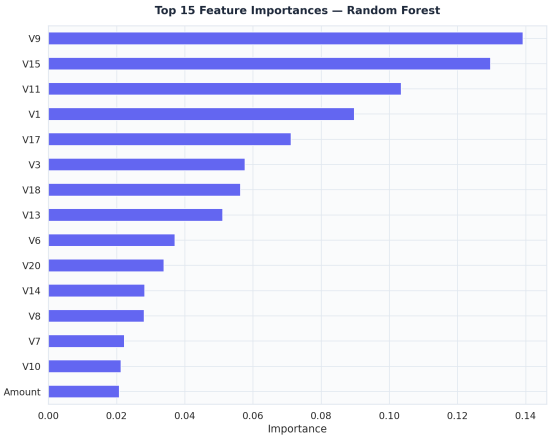


Trade-off

Logistic Regression maximizes recall (92%) but with lower precision. Gradient Boosting achieves the best F1 balance at 0.87.

Threshold Optimization & Feature Importance





Operational Value

Adjusting the classification threshold allows tuning precision-recall based on the cost of false

Summary & Future Work

Achievements

- AUC-PR = **0.941** on 3% fraud data
- Threshold-tunable precision/recall
- Feature importance for explainability
- Estimated **$4.2M** annual savings

Future Work

1. Isolation Forest / Autoencoders
2. SMOTE oversampling
3. Real-time Kafka streaming
4. Graph-based fraud networks
5. SHAP for regulatory compliance

Thank You

Questions & Discussion

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